

Fields 2

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Problem 1.

Proof. Consider rings $A \leq B \leq C$. Suppose C is $f.g$ over B as a B -module and B $f.g$ over A as a A -module. Thus $\exists b_1, \dots, b_n \in B \ni \forall b \in B, b$ is a A -linear combination of $\{b_1, \dots, b_n\}$ or $b = \sum_{k=1}^n a_k b_k$ where $a_k \in A$. Similarly, $\exists c_1, \dots, c_n \in C \ni \forall c \in C, c = \sum_{k=1}^n g_k c_k$ where $g_k \in B$. Thus $c = \sum_{i=1}^n c_i \sum_{j=1}^n a_j b_j = \sum_{i=1}^n \sum_{j=1}^n a_j (b_j c_i)$. Thus c is a A -linear combination of $\{b_j c_i\}_{i,j}$ which is finite since $\{b_j\}_j$ and $\{c_i\}_i$ finite. Thus C is a $f.g$ over A .

The generators of C are the set $\{b_j c_i\}_{i,j}$.

Suppose A is Noetherian and C is finitely generated A module.

1. Case 1: Suppose $C = R^m$. Induct on m .

Base case: Suppose $m = 1$. Every submodule of C is a ideal of A and thus is finitely generated by the definition of Noetherian.

Inductive Hypothesis: Suppose for $m = k$, the theorem is true.

Inductive Step: Suppose $m = k + 1$. Consider the projection map $\varphi : C \rightarrow R^k$ given by dropping the last entry: $\varphi(\alpha_1, \dots, \alpha_m) = (\alpha_1, \dots, \alpha_{m-1})$. $\ker \varphi \cong R$ which is finitely generated. Let W be a submodule of C . $\varphi(W) \subseteq R^{m-1}$ and thus $\varphi(W)$ is finitely generated by induction. $\ker \varphi \cap W$ is a submodule of $\ker \varphi$ and thus is a ideal and thus finitely generated. Thus W is finitely generated.

2. Case 2: The general case: Let C be a general finitely generated R module. Then there is a surjective map from $\varphi : R^m \rightarrow C$. Given a submodule W of C , the correspondence theorem tells us $U = \varphi^{-1}(W)$ is a submodule of R^m and thus finitely generated and thus W is finitely generated.

Since B is submodule of C its finitely generated as A module. Thus C must be finitely generated over B , because otherwise it would be infinitely generated over A which is a contradiction.

Consider rings $A \leq B \leq C$. Suppose C is free with finite rank over B and B free with finite rank over A as a A -module. Thus $\exists b_1, \dots, b_n \in B \ni \forall b \in B, b$ is a A -linear combination of $\{b_1, \dots, b_n\}$ or $b = \sum_{k=1}^n a_k b_k$ where $a_k \in A$. Similarly, $\exists c_1, \dots, c_n \in C \ni \forall c \in C, c = \sum_{k=1}^n g_k c_k$ where $g_k \in B$. Thus $c = \sum_{i=1}^n c_i \sum_{j=1}^n a_j b_j = \sum_{i=1}^n \sum_{j=1}^n a_j (b_j c_i)$. Thus c is a A -linear combination of $\{b_j c_i\}_{i,j}$ which is finite since $\{b_j\}_j$ and $\{c_i\}_i$ finite. Thus C is free over A and $\text{rank}_A C = |\{b_j c_i\}_{i,j}| = \text{rank}_A(B) \text{rank}_B(C)$

□

Problem 2.

Proof. $A \subset B \subset C$ with C finite over A . Since A is a field its only ideal is 0 and thus finitely generated and thus A is Noetherian. Thus B is finite dimensional over A . and C is finite dimensional over B . Since $C \cong B^n$ and $B \cong A^m$ and thus are free modules, $[C : A] = [B : A][C : B]$ $\square$

1 Section I

Problem 1.1.

Proof. $\square$

Problem 1.2.

Proof. Let F be a field, not of characteristic 2, let $x^2 + bx + c = 0$. Suppose $\exists \delta \in F \ni \delta^2 = b^2 - 4c$.

$$\begin{aligned} x^2 + bx + c &= \frac{\delta^2 - 2b\delta + b^2}{4} + \frac{b\delta - b^2}{2} + c \\ &= \frac{1}{4}(\delta^2 - 2b\delta + b^2 + 2bd - 2b^2 + 4c) \\ &= \frac{1}{4}(b^2 - 4c + b^2 - 2b^2 - 4c) = 0 \end{aligned}$$

Suppose, $\nexists \delta \in F \ni \delta^2 = b^2 - 4c$,

Assume the contrary, $x^2 + bx + c = 0$ has a solution.

$$\begin{aligned} x^2 + bx + c &= x^2 + bx + \frac{b^2}{4} + c - \frac{b^2}{4} = 0 \\ (x - b)^2 &= \frac{b^2}{4} - c \\ 4(x - b)^2 &= b^2 - 4c = \delta \\ (x - b)^2 &= \frac{\delta^2}{4} \end{aligned}$$

Since δ doesn't exist we have reached a contradiction. Thus $x^2 + bx + c$ has no solution. $\square$

Problem 1.3.

Proof. $\overline{\mathbb{R}} = \mathbb{R}$, thus subfields of $\mathbb{C}$ which are contained in $\mathbb{R}$ are not dense in $\mathbb{C}$.

Suppose, $\mathbb{K} \subseteq \mathbb{C}$ is a proper subfield of $\mathbb{C}$ such that $\mathbb{K} \not\subseteq \mathbb{R}$. Thus $\exists z_0 \in \mathbb{K} \ni z_0 \notin \mathbb{R}$ thus $z_0 = a + bi$. $\mathbb{K}$ cannot have the reals, otherwise all $\mathbb{K} = \mathbb{C}$. But it must have the rationals and the rational multiples of z_0 .

Let $z \in \mathbb{C}$, we can write $z = c + dz_0$ if $z = x + iy$, let $d = \frac{y}{b}$ and $c = x - \frac{y}{d}a$. If $r, s \in \mathbb{Q}$, then $r + sz_0 \in \mathbb{K}$. By choosing r, s arbitrarily close to a, b we can make $r + sz_0$ as close to z . Thus $\mathbb{K}$ dense in $\mathbb{C}$. $\square$

2 Section II

Problem 2.1.

Proof. $\alpha^3 - 3\alpha + 4 = 0 \implies \alpha^3 = 3\alpha - 4$.

Thus,

$$1(\alpha^2 + \alpha + 1) = \alpha^2 + \alpha + 1$$

$$\begin{aligned}\alpha(\alpha^2 + \alpha + 1) &= \alpha^3 + \alpha^2 + \alpha \\ &= \alpha^2 + 4\alpha - 4\end{aligned}$$

$$\begin{aligned}\alpha^2(\alpha^2 + \alpha + 1) &= \alpha(\alpha^2 + 4\alpha - 4) \\ &= \alpha^3 + 4\alpha^2 - 4\alpha \\ &= 4\alpha^2 + 3\alpha - 4 - 4\alpha \\ &= 4\alpha^2 - \alpha - 4\end{aligned}$$

Thus,

$$\begin{aligned}1 &= (c\alpha^2 + b\alpha + a)(\alpha^2 + \alpha + 1) = 4c\alpha^2 - c\alpha - 4c + b\alpha^2 + 4b\alpha - 4b + a\alpha^2 + a\alpha + a \\ &= \alpha^2(a + b + 4c) + \alpha(a + 4b - c) + (a - 4b - 4c)\end{aligned}$$

Which makes a matrix equation like below

$$\begin{bmatrix} 1 & 1 & 4 \\ 1 & 4 & -1 \\ 1 & -4 & -4 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$a = \frac{17}{49}, b = -\frac{5}{49}, c = -\frac{3}{49}.$$

□

Problem 2.2.

Proof. Let $f(x) = x^n + \sum_{m=0}^m (-1)^{n-m} a_m x^m$. α is a root of f in extension field K .

Thus,

$$\begin{aligned}\alpha^n + \sum_{m=1}^m (-1)^{n-m} a_m \alpha^m + a_0 &= 0 \\ \alpha^n + \sum_{m=1}^m (-1)^{n-m} a_m \alpha^m &= -a_0 \\ \alpha(\alpha^{n-1} + \sum_{m=1}^m (-1)^{n-m} a_m \alpha^{m-1}) &= -a_0 \\ \frac{1}{\alpha} &= -\frac{1}{a_0} \left(\alpha^{n-1} + \sum_{m=1}^m (-1)^{n-m} a_m \alpha^{m-1} \right)\end{aligned}$$

□

Problem 2.3.

Proof. Let $\alpha = \sqrt[3]{2} \omega = e^{2\pi i/3}$ and $\beta := \omega \sqrt[3]{2}$. The minimal polynomial of α is $x^3 - 2$. This is also the minimal polynomial of β . $x^3 - 2$ is irreducible over $\mathbb{Q}$. Thus by Proposition 15.2.8, $\mathbb{Q}[\alpha] \cong \mathbb{Q}[\beta]$.

$$\begin{array}{ccc} \mathbb{Q}[x] (f) & \xleftarrow{\cong} & \mathbb{Q}[\alpha] \\ & \searrow \cong & \uparrow \cong \\ & & \mathbb{Q}[\beta] \end{array}$$

Thus since $x_1^2 + \dots + x_2^2 = -1$ has no solutions in $\mathbb{Q}[\alpha]$ it has no solutions in $\mathbb{Q}[\beta]$. $\square$

Section III

Problem 3.1.

Proof. Let F be a field and let $\deg F[\alpha] = \deg F[\alpha^2] = 5$. $F \subset F[\alpha^2] \subset F[\alpha]$. By the multiplication of degree, $5 = [F[\alpha] : F] = [F[\alpha^2] : F][F[\alpha^2] : F[\alpha]] = 5[F[\alpha^2] : F[\alpha]] \implies [F[\alpha^2] : F[\alpha]] = 1 \implies F[\alpha^2] = F[\alpha]$. $\square$

Problem 3.2.

Proof. Let $f(x) = x^4 + 3x + 3$. By the Eisenstein criterion, $x^4 + 3x + 3 \pmod{3} = x^3$ and $9 \nmid 3$. Thus $x^4 + 3x + 3$ is irreducible in $\mathbb{Q}$. Let r be a generic root of f . $[\mathbb{Q}[r] : \mathbb{Q}] = 4$. $[\mathbb{Q}[\sqrt[3]{2}] : \mathbb{Q}] = 3$. Assume the contrary, $r \in \mathbb{Q}[\sqrt[3]{2}]$. Thus $\mathbb{Q}[r] \subset \mathbb{Q}[\sqrt[3]{2}]$. By the multiplication of degree, $4 = [\mathbb{Q}[\sqrt[3]{2}] : \mathbb{Q}] = [\mathbb{Q}[r] : \mathbb{Q}][\mathbb{Q}[r] : \mathbb{Q}[\sqrt[3]{2}]] = 3[\mathbb{Q}[r] : \mathbb{Q}[\sqrt[3]{2}]]$, however $3 \nmid 4$, thus $[\mathbb{Q}[r] : \mathbb{Q}[\sqrt[3]{2}]]$ is not a positive integer. Thus $\mathbb{Q}[r] \not\subset \mathbb{Q}[\sqrt[3]{2}]$. Thus by contradiction, $r \notin \mathbb{Q}[\sqrt[3]{2}]$ and since r is a generic root of f no root of f is in $\mathbb{Q}[\sqrt[3]{2}]$ and thus f is irreducible over $\mathbb{Q}[\sqrt[3]{2}]$. $\square$

Let $\zeta_n = e^{2\pi i/n}$ for the upcoming problems.

Problem 3.3.

Proof. Assume the contrary that $\zeta_5 \in \mathbb{Q}[\zeta_7]$. Thus $\mathbb{Q}[\zeta_5] \subseteq \mathbb{Q}[\zeta_7]$. $[\mathbb{Q}[\zeta_5] : \mathbb{Q}] = 4$ and $[\mathbb{Q}[\zeta_7] : \mathbb{Q}] = 6$ by the cyclotomic polynomial. By multiplication of degrees, 4 must divide 6 which is a contradiction. Thus by contradiction, $\zeta_5 \notin \mathbb{Q}[\zeta_7]$. $\square$

Problem 3.4a.

Proof. $\zeta_4 = i$

1. The minimal polynomial is $x^2 + 1$.
2. Assume the contrary, $i \in \mathbb{Q}[\zeta_3]$. Then, $\mathbb{Q} \subseteq \mathbb{Q}[i] \subseteq \mathbb{Q}[\zeta_3]$. Since $[\mathbb{Q}[\zeta_3] : \mathbb{Q}] = 2$ and $[\mathbb{Q}[i] : \mathbb{Q}] = 2$. $\mathbb{Q}[i] = \mathbb{Q}[\zeta_3]$. Thus $\zeta_3 = \frac{1}{2} + i\sqrt{3}/2$ can be written as $a + bi$ where $a, b \in \mathbb{Q}$. However $\sqrt{3}/2$ is irrational thus no rational number $b = \sqrt{3}/2$. Thus a contradiction occurs and $i \notin \mathbb{Q}[\zeta_3]$. Thus the $x^2 + 1$ is still minimal over $\mathbb{Q}[\zeta_3]$.

□

Problem 3.4b.

Proof. $\zeta_6 = e^{2\pi i/6} = e^{i\pi/3}$

1. ζ_6 is a root of $x^3 + 1 = (x^2 - x + 1)(x + 1)$. It is not a root of $x + 1$ but is a root of $x^2 - x + 1$. The only possible rational roots of $x^2 - x + 1$ are ± 1 which are not roots of the poly and thus $x^2 - x + 1$ is irreducible and thus is the minimal polynomial of ζ_6 in $\mathbb{Q}$.
2. $\frac{1}{2} + i\sqrt{3}/2 = \zeta_6 = -\zeta_3^2 = -(-\frac{1}{2} - i\sqrt{3}/2)$. Thus ζ_6 is the root of the polynomial $x - \zeta_3^2$. Thus the minimal polynomial of ζ_6 in $\mathbb{Q}[\zeta_3]$ is $x - \zeta_3^2$

□

Problem 3.4c.

Proof. $\zeta_8 = e^{\pi i/4}$

1. ζ_8 is a root to $x^4 + 1$ which has no real roots, as $\forall r \in \mathbb{R}, r^4 > 0$. Thus $x^4 + 1$ is irreducible in $\mathbb{Q}$ and thus the minimal polynomial.
2. The roots of $x^4 + 1$ are $\zeta_8, \zeta_8^3, \zeta_8^5, \zeta_8^7$ whose squares are $\pm i$. By 3.4a, $i \notin \mathbb{Q}[\zeta_3] \implies \zeta_8, \zeta_8^3, \zeta_8^5, \zeta_8^7 \notin \mathbb{Q}[\zeta_3]$, thus $x^4 + 1$ cannot be factored into a linear term. If $x^4 + 1$ is a product of two irreducible quadratic in $\mathbb{Q}[\zeta_3][X]$, it must break into two quadratic terms one of which must have ζ_8 as a root.

$$\text{Thus } (x - \zeta_8)(x - \zeta_8^3) = x^2 - x(\zeta_8 - \zeta_8^3) + \zeta_8^4 = x^2 - xi\sqrt{2} - 1$$

$$(x - \zeta_8)(x - \zeta_8^5) = x^2 - x(\zeta_8 - \zeta_8^5) + \zeta_8^6 = x^2 - i. \quad -i \notin \mathbb{Q}[\zeta_3]. \text{ Thus not a valid quadratic.}$$

$$(x - \zeta_8)(x - \zeta_8^7) = x^2 - x\sqrt{2} + 1$$

Assume the contrary, $\sqrt{2} \in \mathbb{Q}(\sqrt{3}, i)$. Thus $\sqrt{2} = a + bi + c\sqrt{3} + di\sqrt{3} = (a + c\sqrt{3}) + i(b + d\sqrt{3}) \implies b, d = 0$ and $c \neq 0$ because $\sqrt{2} \notin \mathbb{Q}$. Thus $2 = (a + c\sqrt{3})^2 = a^2 + 2ac + 3c^2 \implies 3c^2 = 2 - a^2 - 2ac \implies \sqrt{3} = \frac{2 - a^2 - 2ac}{c} \implies \sqrt{3} \in \mathbb{Q}$. However $\sqrt{3} \notin \mathbb{Q}$, thus by contradiction $\sqrt{2} \notin \mathbb{Q}(\sqrt{3}, i)$.

Assume the contrary, $i\sqrt{2} \in \mathbb{Q}(\sqrt{3}, i)$. Thus $i\sqrt{2} = a + bi + c\sqrt{3} + di\sqrt{3} = (a + c\sqrt{3}) + i(b + d\sqrt{3}) \implies a, c = 0$ and $b \neq 0$ because $\sqrt{2} \notin \mathbb{Q} \implies i\sqrt{2} \notin \mathbb{Q}[i]$. Thus $2 = -(b + d\sqrt{3})^2 = -b^2 - 2bd - 3d^2 \implies -3d^2 = 2 + a^2 + 2ac \implies \sqrt{3} = -\frac{2 + a^2 + 2ac}{c} \implies \sqrt{3} \in \mathbb{Q}$. However $\sqrt{3} \notin \mathbb{Q}$, thus by contradiction $i\sqrt{2} \notin \mathbb{Q}(\sqrt{3}, i)$.

$$\mathbb{Q}(\zeta_8) \subseteq \mathbb{Q}(\sqrt{3}, i) \text{ but } \sqrt{2} \text{ and } i\sqrt{2} \text{ not in } \mathbb{Q}(\sqrt{3}, i), \text{ thus } \sqrt{2}, i\sqrt{2} \notin \mathbb{Q}(\zeta_8)$$

Thus $x^4 + 1$ cannot be reduced and thus is the minimal polynomial for ζ_8

□

Problem 3.4d.

Proof. $\zeta_9 = e^{2\pi i/9}$

1. ζ_9 is a root to $x^9 - 1 = (x^3 - 1)(x^6 + x^3 + 1)$. ζ_9 not a root to $x^3 - 1$. ζ_9 is a root to $f(x) = x^6 + x^3 + 1$. $f(x)$ is irreducible iff $f(x + 1)$ since $x \rightarrow x + 1$ is a automorphism of $\mathbb{Q}[x]$.

Thus,

$$\begin{aligned}
f(x + 1) &= (x + 1)^6 + (x + 1)^3 + 1 \\
&= \sum_{j=0}^6 \binom{6}{j} x^j + \sum_{j=0}^3 \binom{3}{j} x^j + 1 \\
&= x^6 + 6x^5 + 15x^4 + 20x^3 + x^3 + 15x^2 + 3x^2 + 6x + 3x + 1 + 1 + 1 \\
&= x^6 + 6x^5 + 15x^4 + 21x^3 + 18x^2 + 9x + 3
\end{aligned}$$

By the eisenstein criterion, $f(x + 1)$ is irreducible thus $f(x)$ is irreducible.

2. $\mathbb{Q} \subseteq \mathbb{Q}[\zeta_3] \subseteq \mathbb{Q}[\zeta_9]$. $\zeta_9^3 = \zeta_3$. $6 = [\mathbb{Q}[\zeta_9] : \mathbb{Q}] = [\mathbb{Q}[\zeta_9] : \mathbb{Q}[\zeta_3]][\mathbb{Q}[\zeta_3] : \mathbb{Q}] = 2[\mathbb{Q}[\zeta_9] : \mathbb{Q}[\zeta_3]] \implies [\mathbb{Q}[\zeta_9] : \mathbb{Q}[\zeta_3]] = 3 \implies$ the minimal polynomial is $x^3 - \zeta_3$.

□

Problem 3.4e.

Proof. $\zeta_{10} = e^{\pi i/5}$

1. ζ_{10} is a root of $x^5 + 1 = (x - 1)(x^4 - x^3 + x^2 - x + 1)$. ζ_{10} is not a root of $x - 1$ but is a root of $f(x) = x^4 - x^3 + x^2 - x + 1$. Thus $[\mathbb{Q}[\zeta_{10}] : \mathbb{Q}] \leq 4$. $\zeta_{10}^2 = \zeta_5$. Thus $\mathbb{Q}[\zeta_5] \subseteq \mathbb{Q}[\zeta_{10}]$. ζ_5 is the solution of the polynomial $f(x) = x^4 + x^3 + x^2 + 1$.

$$\begin{aligned}
f(x + 1) &= x^4 + 4x^3 + x^3 + 6x^2 + 3x^2 + x^2 + x + 4x + 3x + 2x + 5 \\
&= x^4 + 5x^3 + 10x^2 + 10x + 5
\end{aligned}$$

$f(x + 1)$ is irreducible by eisenstein criterion. Thus $f(x)$ is irreducible.

But $[\mathbb{Q}[\zeta_5] : \mathbb{Q}] = 4$. Since $[\mathbb{Q}[\zeta_{10}] : \mathbb{Q}] = 4[\mathbb{Q}[\zeta_{10}] : \mathbb{Q}[\zeta_5]] \leq 4 \implies [\mathbb{Q}[\zeta_{10}] : \mathbb{Q}] = 4$ and $\mathbb{Q}[\zeta_5] = \mathbb{Q}[\zeta_{10}]$

2. Since $\mathbb{Q}[\zeta_5] = \mathbb{Q}[\zeta_{10}]$, it suffices to show that the minimal polynomial for ζ_5 is not reducible over $\mathbb{Q}$. Assume the contrary, $f(x)$ is reducible over $\mathbb{Q}[\zeta_3] \implies \mathbb{Q} \subseteq \mathbb{Q}[\zeta_3] \subseteq \mathbb{Q}[\zeta_5]$. Since all the intermediate fields of $\mathbb{Q}[\zeta_5]$ can be made by taking all the g such that $g|f$ and adjoining all the coefficients of g to $\mathbb{Q}$. The only roots of f are $\zeta_5, \dots, \zeta_5^4$ and $[\mathbb{Q}[\zeta_5] : \mathbb{Q}[\zeta_3]] = \frac{4}{2} = 2$. So we have to take all possible pairs to see if all the coefficients are $\mathbb{Q}[\zeta_3]$.

Thus $(x - \zeta_5^i)(x - \zeta_5^j) = x^2 + (\zeta_5^i + \zeta_5^j)x + \zeta_5^{i+j}$.

If $j = 5 - i$, we get $x^2 + 2x \cos(\frac{2\pi i}{5}) + 1$. $2x \cos(\frac{2\pi i}{5}) \notin \mathbb{Q}[\zeta_3]$.

If $j \neq 5 - i$, we get $x^2 + 2x(\zeta_5^i + \zeta_5^j) + \zeta_5^{i+j}$. But ζ_5^{i+j} generates all of μ_5 . Thus we get $\mathbb{Q}[\zeta_5]$ back which is not in $\mathbb{Q}[\zeta_3]$.

Thus $x^4 + x^3 + x^2 + x + 1$ is irreducible so $x^4 - x^3 + x^2 - x + 1$ is irreducible over $\mathbb{Q}[\zeta_3]$

□

Problem 3.4f.*Proof.* $\zeta_{12} = e^{\pi i/6}$

1. ζ_{12} is a root of $x^6 + 1 = (x + 1)^2(x^4 + x^2 + 1)$. ζ_{12} is not a root of $x - 1$ but is a root of $f(x) = x^4 + x^2 + 1$. $f(x)$ is irreducible because it is the 12th cyclotomic polynomial. Thus f is the minimal polynomial.
2. $\mathbb{Q} \subseteq \mathbb{Q}[\zeta_3] \subseteq \mathbb{Q}[\zeta_{12}]$. $\zeta_{12}^2 = e^{\pi i/3} = -\zeta_3^2$. $[\mathbb{Q}[\zeta_{12} : \mathbb{Q}[\zeta_3]]] = \frac{[\mathbb{Q}[\zeta_{12} : \mathbb{Q}]}{[\mathbb{Q}[\zeta_3 : \mathbb{Q}]} = \frac{4}{2} = 2$. Thus the minimal polynomial is $x^2 + \zeta_3^2$

□

Problem 3.5.

Proof. By the cyclotomic polynomial, we know $[\mathbb{Q}[\zeta_n] : \mathbb{Q}] = \varphi(n)$, or the number of relative numbers less than it. The only numbers with $\varphi(n) \leq 3$ are 2, 3, 4, 6. □

Problem 3.6.

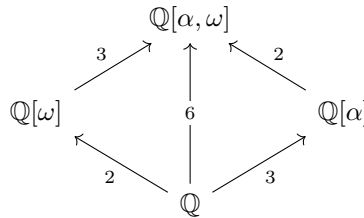
Proof. Let $a \in \mathbb{Q}$ such that it is not a square in $\mathbb{Q}$. $\mathbb{Q} \subseteq \mathbb{Q}[\sqrt{a}] \subseteq \mathbb{Q}[\sqrt[4]{a}]$ and $[\mathbb{Q}[\sqrt{a}] : \mathbb{Q}] = 2$. Assume the contrary, $\sqrt[4]{a} \in \mathbb{Q}[\sqrt{a}] \implies \exists x, y \in \mathbb{Q} \ni \sqrt[4]{a} = x + y\sqrt{a}$. $\sqrt{a} = (x + y\sqrt{a})^2 = x^2 + 2xy\sqrt{a} + y^2a = (x^2 + y^2a) + 2xy\sqrt{a} \implies x^2 + y^2a = 0$. But since $x^2, y^2, a > 0$, $x = 0$, $y = 0$. Which is a contradiction. Thus by contradiction, $\sqrt[4]{a} \notin \mathbb{Q}[\sqrt{a}]$. But $\sqrt[4]{a}$ minimal polynomial in $\mathbb{Q}[\sqrt{a}]$ is $x^2 - \sqrt{a}$. Thus $[\mathbb{Q}[\sqrt[4]{a}] : \mathbb{Q}[\sqrt{a}]] = 2$. Thus $[\mathbb{Q}[\sqrt[4]{a}] : \mathbb{Q}] = 2 * 2 = 4$. □

Problem 3.7a.

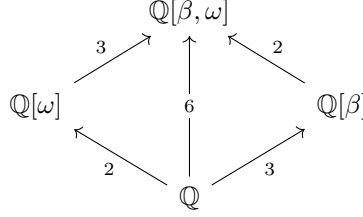
Proof. Let $\alpha = \sqrt[4]{-2}$. $\mathbb{Q} \subseteq \mathbb{Q}[\sqrt{-2}] \subseteq \mathbb{Q}[\alpha]$. Assume the contrary, $i \in \mathbb{Q}[\alpha]$. Thus, $\mathbb{Q} \subseteq \mathbb{Q}[i] \subseteq \mathbb{Q}[\alpha]$. $\sqrt{-2} = i\sqrt{2} \notin \mathbb{Q}[i]$. Thus $\mathbb{Q}[i] \subset \mathbb{Q}[i, \sqrt{-2}] \subseteq \mathbb{Q}[\alpha]$. $[\mathbb{Q}[\alpha] : \mathbb{Q}[i]] = [\mathbb{Q}[\alpha] : \mathbb{Q}] / [\mathbb{Q}[i] : \mathbb{Q}] = \frac{4}{2} = 2$. Since 2 is prime, $\mathbb{Q}[i, \sqrt{-2}] = \mathbb{Q}[\alpha]$. Thus $\alpha = a + b\sqrt{-2}$ for $a, b \in \mathbb{Q}[i]$. $\alpha^2 = \sqrt{-2} = a^2 + 2ab\sqrt{-2} - 2b^2 = (a^2 - 2b^2) + 2ab\sqrt{-2}$. Thus $0 = a^2 - 2b^2 = (a - \sqrt{2}b)(a + \sqrt{2}b)$ this means, $a = \sqrt{2}b$ or $a = -\sqrt{2}b$ either of which is impossible to satisfy if $a, b \in \mathbb{Q}[i]$. $a = b = 0$ is also a valid solution but this implies $\sqrt{-2} = 0$ which is a contradiction. Thus by contradiction, $i \notin \mathbb{Q}[\alpha]$. □

Problem 3.7b.

Proof. Let $\omega = \zeta_3$, $\alpha = \sqrt[3]{2}$, and $\beta = \sqrt[3]{5}$. It suffices to prove that $\beta \notin \mathbb{Q}[\alpha, \omega]$ because, $\mathbb{Q}[\beta] \subset \mathbb{Q}[\alpha, \omega]$. Since 3 and 2 are coprime,



and



Assume the contrary, $\mathbb{Q}[\beta, \omega] = \mathbb{Q}[\alpha, \omega]$. $\mathbb{Q}[\alpha, \omega]$ is generated by two automorphisms σ which interchanges ω and ω^2 and γ which sends $\alpha \rightarrow \omega\alpha$ and sends $\omega\alpha \rightarrow \omega^2\alpha$ but fixes ω and ω^2 . γ can't fix β or it would fix the whole real subfield of $\mathbb{Q}[\alpha, \beta]$ and it already fixes ω and ω^2 , so it would be trivial. But it must send β to another cube root of 5.

Suppose, $\gamma(\beta) = \omega\beta$. Then γ and σ fix $\frac{\beta}{\alpha}$ so $\frac{\beta}{\alpha} \in \mathbb{Q} \implies \sqrt[3]{\frac{5}{2}} \in \mathbb{Q}$. Thus $\exists a, b \in \mathbb{Z} \ni \gcd(a, b) = 1 \ni \frac{a}{b} = \frac{\beta}{\alpha} \implies \frac{a^3}{b^3} = \frac{5}{2} \implies 2a^3 = 5b^3$ thus $5|a \implies 5|b$ which is a contradiction.

Suppose then $\gamma(\beta) = \omega^2\beta$. Then $\gamma(\beta^2) = \omega\beta^2$. Thus γ and σ both fix $\frac{\beta^2}{\alpha} \implies \frac{\beta^2}{\alpha} \in \mathbb{Q}$. By the same argument as above this is a contradiction.

Thus $\beta \in \mathbb{Q}[\alpha, \omega]$ produces a contradiction. Thus by contradiction, $\beta \notin \mathbb{Q}[\alpha, \omega] \implies \beta \notin \mathbb{Q}[\alpha]$. $\square$

Problem 3.8.

Proof. Let $\alpha, \beta \in \mathbb{C}$. Suppose $\alpha + \beta$ and $\alpha\beta$ are algebraic numbers. Thus $\mathbb{Q}[\alpha + \beta, \alpha\beta]$ is finitely dimensional wrt to $\mathbb{Q}$. Take $(x - \alpha)(x - \beta) = x^2 - (\alpha + \beta)x + \alpha\beta \in \mathbb{Q}[\alpha + \beta, \alpha\beta][X]$. α and β are roots of the polynomial. Thus $\mathbb{Q}[\alpha, \beta]$ is a finite dimensional with respect to $\mathbb{Q}[\alpha + \beta, \alpha\beta]$. Thus $\mathbb{Q}[\alpha, \beta]$ is finite dimensional with respect to $\mathbb{Q}$. Since $\mathbb{Q}[\alpha] \subseteq \mathbb{Q}[\alpha, \beta]$ and $\mathbb{Q}[\beta] \subseteq \mathbb{Q}[\alpha, \beta]$ they are also finite dimensional wrt to $\mathbb{Q}$ and thus α and β are algebraic. $\square$

Problem 3.9.

Proof. Let $K = \mathbb{Q}[\alpha]$ and $L = \mathbb{Q}[\beta]$. Let $m = \deg f$ and $n = \deg g$. Suppose $f(x)$ is irreducible over $L[x]$. It is also irreducible over $\mathbb{Q}$. Thus $K \cong \mathbb{Q}[X]/f(x) \implies [K, \mathbb{Q}] = m$. Similarly, $\mathbb{Q}[\alpha, \beta] \cong L[X]/f(x) \implies [\mathbb{Q}[\alpha, \beta] : L] = m$. $L \cong \mathbb{Q}[x]/g$ since g is a irreducible polynomial where β . Thus $[L : \mathbb{Q}] = n$. Thus since $[Q(\alpha, \beta) : \mathbb{Q}] = [L : \mathbb{Q}][\mathbb{Q}(\alpha, \beta) : L] = mn$. Thus $[\mathbb{Q}[\alpha, \beta] : K] = n \implies g$ is irreducible over $K[X]$. $\square$

Problem 3.10. K/F and L/K algebraic $\implies L/F$ algebraic

Proof. $\forall l \in L, \exists g \in K[X] \ni g(l) = 0$. Let $g(x) = \sum_{j=0}^n k_j x^j$ where $k_j \in K$.

$$[F[l, k_0, \dots, k_n] : F] = [F[l, k_0, \dots, k_n] : F[k_0, \dots, k_n]][F[k_0, \dots, k_n] : F]$$

$[F[l, k_0, \dots, k_n] : F[k_0, \dots, k_n]]$ is finite because $g \in F[k_0, \dots, k_n][X]$ and $g(l) = 0$. $[F[k_0, \dots, k_n] : F]$ is finite because k_j is algebraic thus $\exists h_j \in F[X] \ni h_j(k_j) = 0$. Let $h(x_1, \dots, x_n) = \sum_{i=0}^n h_i(x_i)$ and $(k_0, \dots, k_n)$ is a root of h .

Thus $[F[l, k_0, \dots, k_n] : F]$ is finite. Because $F \subseteq F[l] \subseteq [F[l, k_0, \dots, k_n], [F[l] : F]] \leq [F[l, k_0, \dots, k_n] : F] \implies [F[l] : F]$ is finite and thus l is algebraic over F .

Thus L/F is algebraic. $\square$